

Negative Exponential Distribution:

CLASSMATE

Date _____

Page _____

A random variable X is said to be followed a negative exponential distribution with parameter θ if its probability density function (pdf) is defined as,

$$f(x) = \theta e^{-\theta x}, \quad x > 0 \\ = 0, \quad \text{otherwise}$$

Moments of NED:

The r th moment about origin of negative exponential distribution is given by,

$$\mu_r' = E(X^r) = \int_0^{\infty} x^r f(x) dx$$

$$= \int_0^{\infty} x^r \theta e^{-\theta x} dx$$

$$\text{Put } y = \theta x \Rightarrow \boxed{x = y/\theta}$$

$$dy = \theta dx$$

$$\boxed{dx = \frac{dy}{\theta}}$$

$$\mu_r' = \theta \int_0^{\infty} (y/\theta)^r e^{-y} \frac{dy}{\theta}$$

$$= \frac{1}{\theta^r} \int_0^{\infty} y^r e^{-y} dy$$

$$= \frac{1}{\theta^r} \left[\int_0^{\infty} y^{(r+1)-1} e^{-y} dy \right]$$

$$= \frac{1}{\theta^r} \Gamma(r+1)$$

$$\mu_r' = \frac{r!}{\theta^r}$$

$$\text{Put } r = 1, 2, 3, 4$$

Therefore,

$$\boxed{\mu_1' = \frac{1}{\theta}}$$

$$\mu_2' = \frac{2!}{\theta^2} = \frac{2}{\theta^2}$$

$$\mu_3' = \frac{3!}{\theta^3} = \frac{6}{\theta^3}$$

$$\mu_4' = \frac{4!}{\theta^4} = \frac{24}{\theta^4}$$

Moments about mean:

$$\Rightarrow \mu_1 = 0$$

$$\Rightarrow \mu_2 = \mu_2' - (\mu_1')^2 = \frac{2}{\theta^2} - \left(\frac{1}{\theta}\right)^2 = \frac{2-1}{\theta^2}$$

$$V(x) \left| \mu_2 = \frac{1}{\theta^2} \right|$$

$$\Rightarrow \mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$= \frac{2}{\theta^3}$$

$$\Rightarrow \mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

$$= \frac{9}{\theta^4}$$

Page _____

$$\text{Skewness } (\beta_1) = \frac{\mu_3^2}{\mu_2^3} = \frac{(2/10^3)^2}{(4/10^2)^3} = 9$$

$$\text{Kurtosis } (\beta_2) = \frac{\mu_4}{\mu_2^2} = \frac{9/10^4}{(4/10^2)^2} = 9$$

Ans: $\rho = -0.46$

$$\gamma_1 = \sqrt{\beta_1} = \sqrt{9} = 3$$

> 0

$$\gamma_2 = \beta_2 - 3 = 9 - 3 = 6$$

Moment Generating Function of NED:

Let x be a r.v. with pdf

$$f(x) = \theta e^{-\theta x}, \quad x > 0$$

The mgf of x is given by,

$$M_x(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} f(x) dx$$

$$= \int_0^{\infty} e^{tx} \theta e^{-\theta x} dx$$

$$= \theta \int_0^{\infty} e^{-(\theta-t)x} dx$$

$$= \frac{\theta}{\theta-t} \left[e^{-(\theta-t)x} \right]_0^{\infty}$$

$$= \frac{\theta}{\theta-t} = \frac{\theta}{\theta(1-t/\theta)}$$

$$= \left(1 - \frac{t}{\theta}\right)^{-1}$$

$$\therefore M_x(t) = \left(1 - \frac{t}{\theta}\right)^{-1}$$

Expanding,

$$M_x(t) = 1 + \frac{t}{\theta} + \frac{t^2}{\theta^2} + \frac{t^3}{\theta^3} + \dots + \frac{t^r}{\theta^r}$$

$\mu_r' = \text{coefficient of } \frac{t^r}{r!} \text{ in } M_X(t)$

Interquartile Range of Negative Exponential Distribution:

The first quartile Q_1 is given by,

$$\int_0^{Q_1} f(x) dx = 1/4$$

$$\text{or, } \int_0^{Q_1} \theta e^{-\theta x} dx = 1/4$$

$$\theta \int_0^{Q_1} e^{-\theta x} dx = 1/4$$

$$\frac{\theta}{\theta} \left[\frac{e^{-\theta x}}{-\theta} \right]_0^{Q_1} = 1/4$$

$$\left[-e^{-\theta x} \right]_0^{Q_1} = 1/4$$

$$1 - e^{-\theta Q_1} = 1/4$$

$$e^{-\theta Q_1} = 3/4$$

$$-\theta Q_1 = \log 3/4$$

$$\therefore Q_1 = -\frac{1}{\theta} [\log_e 3 - \log_e 4]$$

Similarly, the third quartile is given by,

$$\int_0^{Q_3} \theta e^{-\theta x} dx = 3/4$$

$$-\theta Q_3 = \ln 1/4$$

$$Q_3 = -\frac{1}{\theta} \ln 1/4$$

$$Q_3 = \frac{1}{\theta} \ln 4$$

The Interquartile range is given by

$$Q_3 - Q_1 = \frac{1}{\theta} \ln 9 - \left[-\frac{1}{\theta} (\ln 3 - \ln 9) \right]$$

$$Q_3 - Q_1 = \frac{1}{\theta} \log_e 3$$

Uses:

- (i) It has a greater application in queuing theory and reliability theory.
- (ii) It is most useful model to study the waiting time until the first change of a certain electronic equipment & other machines or systems.
- (iii) It is most useful distribution in studying the decay time of radioactive particles.

Properties of Negative Exponential Distribution

- (i) Mean, $E(X) = 1/\theta$, variance, $V(X) = 1/\theta^2$
- (ii) The r^{th} moment about origin of NED is given as,

$$\mu_r' = \frac{r!}{\theta^r}, \quad r = 1, 2, 3, 4$$

- (iii) The first four central moment of NED $\text{Exp}(\theta)$ are,

$$\mu_1 = 0, \mu_2 = 1/\theta^2, \mu_3 = 2/\theta^3, \mu_4 = 9/\theta^4$$

- (iv) In negative exponential distribution,

if $\theta = 1$, Mean = Variance

if $\theta > 1$, Mean > Variance

if $\theta < 1$, Mean < Variance

- (v) For Negative exponential distribution,

$$\beta_1 (\text{skewness}) = 3 \Rightarrow \gamma_1 = 2$$

$$\beta_2 (\text{kurtosis}) = 9 \Rightarrow \gamma_2 = 6$$

Thus, the negative exponential distribution is positively skewed and leptokurtic.

(i) Mgf, $M_X(t) = \left(1 - \frac{t}{\theta}\right)^{-1}, t < 0$

(ii) Characteristic function $X \sim \text{Expo}(\theta)$ is,
 $\phi_X(t) = \left(1 - \frac{it}{\theta}\right)^{-1}$

(iii) Interquantile range = $\frac{1}{\theta} \log_e 3$

(iv) NED also possesses the memoryless property.

practical

Fitting of Negative Exponential Distribution:

The distribution of time (in second) to decay 100 identical radioactive particles is given below.

Decay time:	0-0.5	0.5-1.0	1.0-1.5	1.5-2.0	2.0-2.5
no. of particles:	49	26	20	9	1

Fit a negative expo. distribution to the above data and also calculate expected frequencies.

solⁿ

Decay time (X)	no. of particles (f)	midvalue (x)	f x
0-0.5	49	0.25	12.25
0.5-1.0	26	0.75	19.5
1.0-1.5	20	1.25	25
1.5-2.0	9	1.75	15.75
2.0-2.5	1	2.25	2.25
N = 100		$\Sigma fx = 73.5$	

$$\bar{X} = \frac{\Sigma fx}{n} = \frac{73.5}{100} = 0.735$$

Here, $X \sim \text{NED}, (\theta)$

$$E(X) = \frac{1}{\theta} \Rightarrow \bar{x} = \frac{1}{\theta}$$

$$\theta = \frac{1}{\bar{x}}$$

$$\theta = \frac{1}{0.735} = 1.36$$

Now,

$$P(0 < X < 0.5) = \int_0^{0.5} f(x) dx$$

$$= \int_0^{0.5} \theta e^{-\theta x} dx$$

$$= \int_0^{0.5} 1.36 e^{-1.36x} dx$$

$$= \left[-e^{-1.36x} \right]_0^{0.5}$$

$$= 1 - e^{-0.68} = 1 - 0.5066$$

$$[P(0 < X < 0.5) = 0.4934]$$

$$P(0.5 < X < 1) = \int_{0.5}^1 f(x) dx$$

classmate

Date _____

Page _____

$p(a < x < b)$	Expected frequency $= N \times p(a < x < b)$
0.4934	$49.34 \approx 49$
0.2500	$25.0 \approx 25$
0.1266	$12.66 \approx 13$
0.0642	$6.42 \approx 7$
0.0325	$3.25 \approx 3$
	<u>97</u>

Q Example: For the following distribution

$$f(x) = y_0 e^{-x/\sigma} dx, \quad x \geq 0, \quad \sigma > 0$$

- (i) Determine the constant y_0
- (ii) Find the mgf of the distribution and hence show that the mean and standard deviation is equal to σ .
- (iii) Also, show that $B_1 = 4$ and $B_2 = 9$.

Solution:

(i) we have,

$$\int_0^{\infty} f(x) dx = 1$$

$$\text{or, } \int_0^{\infty} y_0 e^{-x/\sigma} dx = 1$$

$$\text{or, } y_0 \int_0^{\infty} e^{-x/\sigma} dx = 1$$

$$\text{or, } y_0 \left[\frac{e^{-x/\sigma}}{-1/\sigma} \right]_0^{\infty} = 1$$

$$\text{or, } y_0 (-\sigma) [0 - 1] = 1$$

$$y_0 = 1/\sigma$$

(17) The mgf is given by,

$$M_X(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} f(x) dx$$

$$= \frac{1}{\sigma} \int_0^{\infty} e^{tx} e^{-x/\sigma} dx$$

$$= \frac{1}{\sigma} \int_0^{\infty} e^{-(\frac{1}{\sigma}-t)x} dx$$

$$= \frac{1}{\sigma} \left[\frac{-e^{-(\frac{1}{\sigma}-t)x}}{(\frac{1}{\sigma}-t)} \right]_0^{\infty}$$

$$= \frac{1}{\sigma} \frac{1}{(\frac{1}{\sigma}-t)} = \frac{1}{(1-\sigma t)}$$

$$= (1-\sigma t)^{-1}, \quad t < \frac{1}{\sigma}$$

$$M_X(t) = 1 + \sigma t + \sigma^2 t^2 + \dots + \sigma^r t^r + \dots$$

The r th moment about origin of the distribution is given by,

$$\mu_r' = \text{coefficient of } \frac{t^r}{r!} \text{ in } M_X(t)$$

$$\mu_1' = 1! \sigma^1$$

$$\therefore \mu_1' = \sigma, \quad \mu_2' = 2! \sigma^2 = 2\sigma^2$$

$$\mu_3' = 3! \sigma^3 = 6\sigma^3$$

$$\mu_4' = 4! \sigma^4 = 24\sigma^4$$

$$\Rightarrow \mu_2 = \mu_2' - \mu_1'^2 = 2\sigma^2 - \sigma^2 = \sigma^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3 = 2\sigma^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4 = 9\sigma^4$$

$$\therefore V(X) = \sigma^2, \quad \text{s.d.}(X) = \sigma$$

$$\text{Hence, Mean} = \text{s.d.} = \sigma$$

$$(18) B_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(2\sigma^3)^2}{(\sigma^2)^3} = 4$$

$$B_2 = \frac{\mu_4}{\mu_2^2} = \frac{9\sigma^4}{(\sigma^2)^2} = 9$$

Imp Memoryless Property of Negative Exponential Distribution.

For any two variable s and t

$$P[X \geq s+t | X \geq s] = P(X \geq t)$$

where X is any random variable whose pdf is given as, $f(x) = \theta e^{-\theta x}$

LHS,

$$P[X \geq s+t | X \geq s] = \frac{P(X \geq s+t \cap X \geq s)}{P(X \geq s)}$$

$$= \frac{P(X \geq s+t)}{P(X \geq s)}$$

$$= \frac{\int_{s+t}^{\infty} f(x) dx}{\int_s^{\infty} f(x) dx}$$

$$= \frac{\int_{s+t}^{\infty} \theta e^{-\theta x} dx}{\int_s^{\infty} \theta e^{-\theta x} dx}$$

$$= \frac{\theta \left[\frac{e^{-\theta x}}{-\theta} \right]_{s+t}^{\infty}}{\theta \left[\frac{e^{-\theta x}}{-\theta} \right]_s^{\infty}}$$

$$= \frac{e^{-\theta s - \theta t}}{e^{-\theta s}} = \frac{e^{-\theta s} \cdot e^{-\theta t}}{e^{-\theta s}}$$

$$= e^{-\theta t}$$

$$\therefore P[X \geq s+t | X \geq s] = e^{-\theta t}$$

$$\text{RHS, } P(X \geq t) = \int_t^{\infty} f(x) dx = \int_t^{\infty} \theta e^{-\theta x} dx$$

$$= \left[-e^{-\theta x} \right]_t^{\infty}$$

$$\boxed{P(X \geq t) = e^{-t\theta}}$$

That means, Negative exponential distribution follow memoryless property.

Example: Let X has an exponential distribution with pdf

$$f(x) = \frac{1}{4} e^{-x/4}, \text{ if } x \geq 0$$

$$= 0, \text{ otherwise}$$

- (i) What is the prob. that X is not less than 5?
- (ii) Find the mean and variance of the distribution.
- (iii) Obtain the mgf of the distribution.

Solⁿ, $f(x) = \frac{1}{4} e^{-x/4}$

Comparing with NED, $f(x) = \theta e^{-\theta x}, x > 0$
where $\theta = 1/4$

Now,

$$(i) P(X < 4) = \int_0^4 f(x) dx = \int_0^4 \frac{1}{4} e^{-x/4} dx$$

$$= \frac{1}{4} \left[\frac{e^{-x/4}}{-1/4} \right]_0^4$$

$$= \left[-e^{-x/4} \right]_0^4 = -0.36 + 1$$

$$= 0.64$$

(iv) Since the distribution is negative exponential distribution, the mean and variance is given by,

$$\text{Mean}[E(X)] = \frac{1}{\theta} = \frac{1}{1/4} = 4$$

$$\text{Variance, } V(X) = \frac{1}{\theta^2} = \frac{1}{(1/4)^2} = 16$$

(11) $\Rightarrow$

Let x be a random variable with pdf,

$$f(x) = \frac{1}{4} e^{-x/4}, \quad x > 0$$

The mgf of x is given by,

$$M_X(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} \cdot \frac{1}{4} e^{-x/4} dx$$

$$= \frac{1}{4} \int_0^{\infty} e^{-(\frac{1}{4}-t)x} dx$$

$$= \frac{1}{4} \left[\frac{e^{-(\frac{1}{4}-t)x}}{-(\frac{1}{4}-t)} \right]_0^{\infty} dx$$

$$= \frac{1}{4} \left[-e^{-(\frac{1}{4}-t)\infty} + e^{-(\frac{1}{4}-t)0} \right]$$

$$= \left(1 - \frac{t}{1/4} \right)^{-1}$$

$$\therefore M_X(t) = \left(1 - \frac{t}{1/4} \right)^{-1}$$

This is the mgf of the given distribution.

Moments of NED:

The r th moment about origin of NED is given by,

$$\mu_r' = E(x^r) = \int_0^{\infty} x^r f(x) dx$$

$$= \int_0^{\infty} x^r \cdot \frac{1}{4} e^{-x/4} dx$$

$$= \frac{1}{4} \int_0^{\infty} x^r e^{-x/4} dx$$

put $y = x/4$

$$\Rightarrow x = 4y$$

$$dx = 4dy$$

$$\Rightarrow \mu_r' = \frac{1}{4} \int_0^{\infty} (4y)^r \cdot e^{-y} \cdot 4dy$$

$$= 4^r \int_0^{\infty} y^r e^{-y} dy$$

$$= 4^r \int_0^{\infty} e^{-y} y^{(r+1)-1} dy$$

$$= 4^r \cdot \Gamma(r+1)$$

$$\mu_r' = 4^r \cdot r!$$

put $r = 1, 2$

$$\mu_1' = 4^1 \cdot 1! = 4$$

$$\Rightarrow E(x) = 4$$

$$\mu_2' = 4^2 \cdot 2! = 16 \times 2 = 32$$

$$V(x) = \mu_2' - \mu_1'^2 = 32 - 4^2 = 32 - 16 = 16$$